Proofs of Quantum Memory

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Motivation

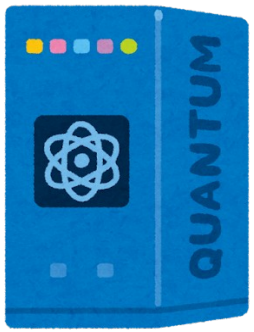
I want to verify it...

We achieve 100-million quantum memory!





classical communication



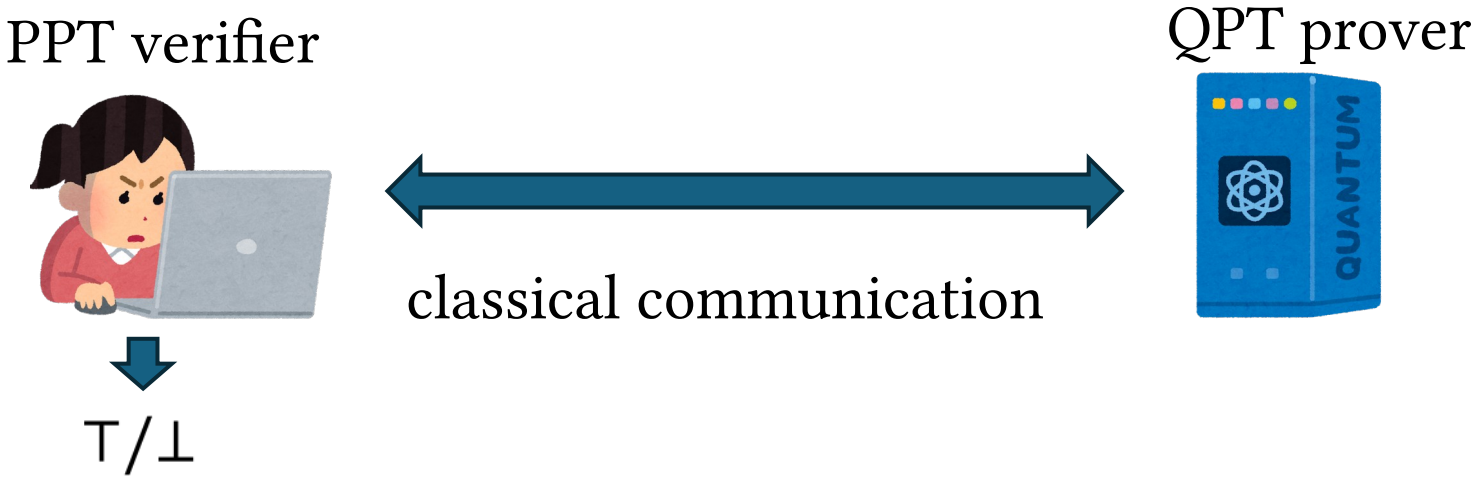
classical verifier

company

Can a classical verifier verify that the company possesses a certain number of qubits for a specified period of time?

Proofs of Quantumness (PoQ) [Brakerski et al'21]

Are Proofs of Quantumness sufficient?



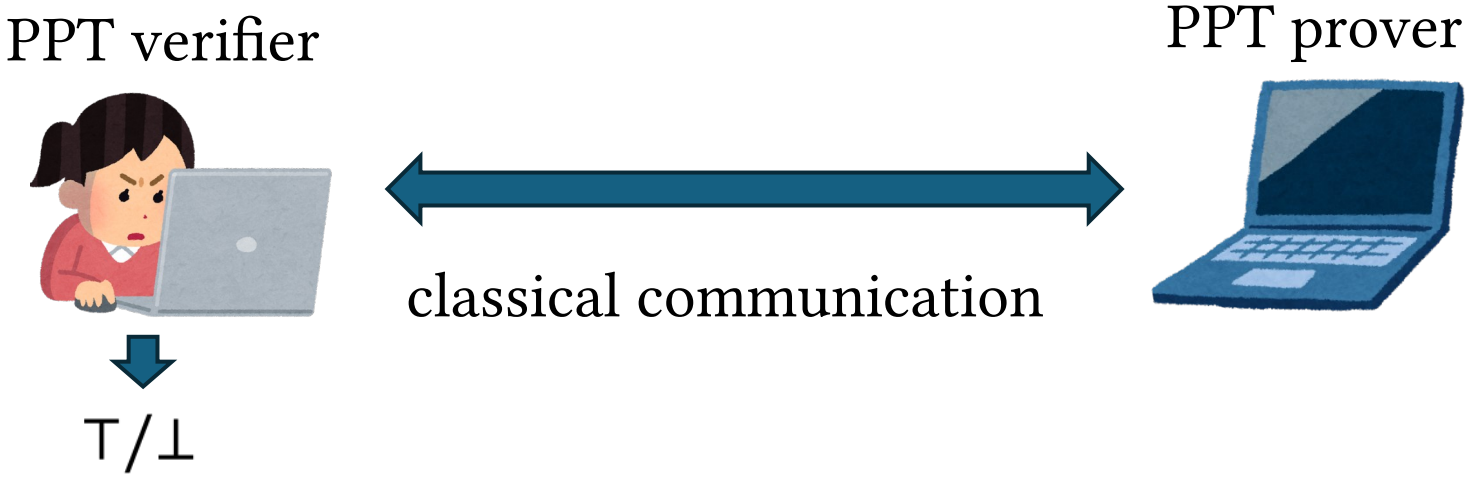
Completeness:
If the QPT prover behaves honestly, the verifier accepts with high probability.

Soundness:
For any PPT prover, the verifier accepts with low probability.

Proofs of Quantumness (PoQ) [Brakerski et al'21]

Are Proofs of Quantumness sufficient?

$\Longrightarrow$ No! They only verify that the prover is not classical .



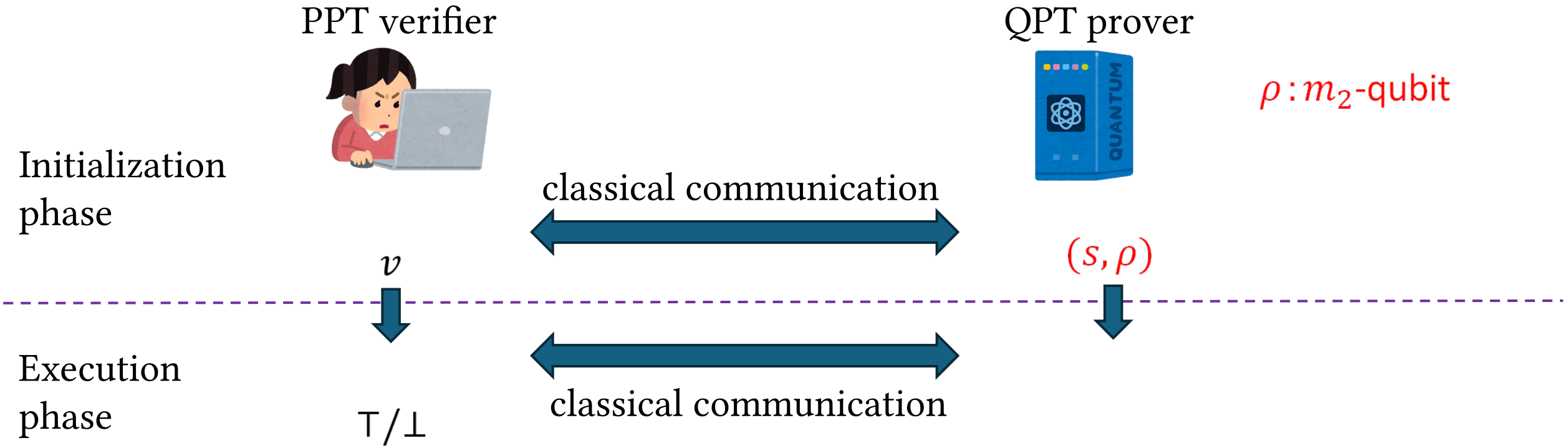
Completeness:
If the QPT prover behaves honestly, the verifier accepts with high probability.

Soundness:
For any PPT prover, the verifier accepts with low probability.

Our Results

- Introducing Proofs of Quantum Memory (PoQM)
- Constructing PoQM based on the hardness of LWE
- Lower bounds for PoQM

Proofs of Quantum Memory (PoQM)



- completeness : If the prover behaves honestly, the verifier accepts with high probability.
- soundness : For any prover that has at most m_2-qubit quantum memory, the verifier accepts with low probability.

the verifier accepts with high probability
⇒ the prover has at least $(m_2 + 1)$-qubit quantum memory.

PoQM construction based on LWE

We present two constructions of PoQM.

more rounds

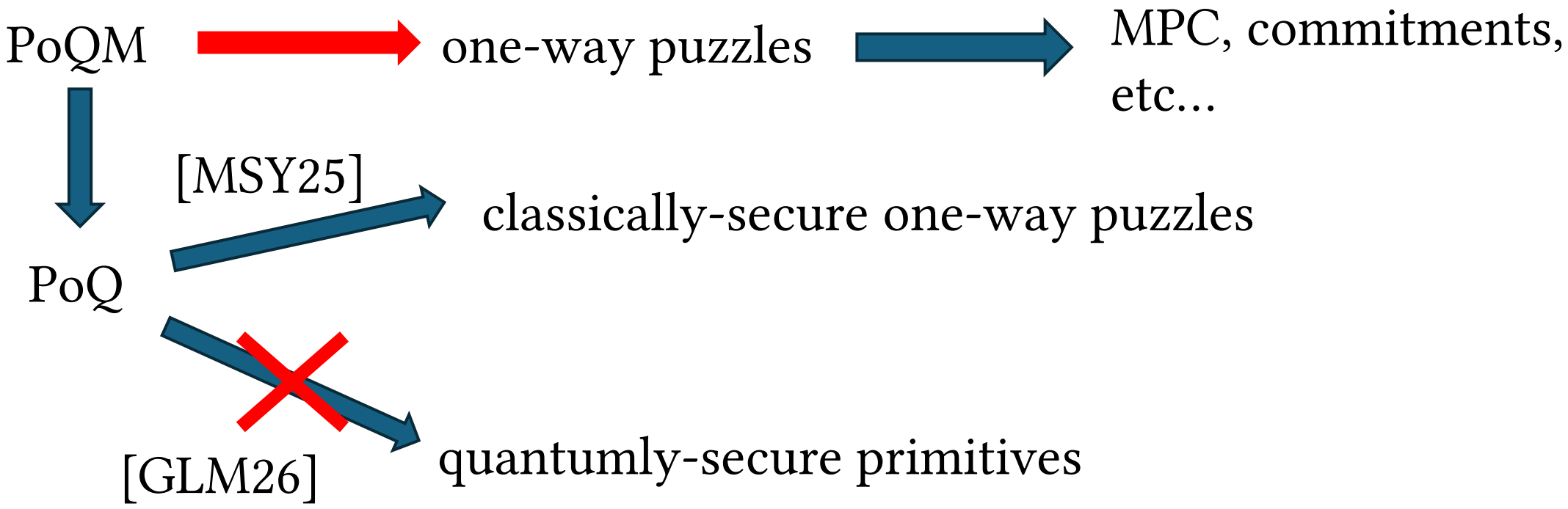
- polynomial hardness of LWE ⇒ $poly$-round
- sub-exponential hardness of LWE ⇒ 2-round

stronger assumption

Lower bounds for PoQM

quantum analogue of one-way functions

PoQM imply quantumly-secure primitive (one-way puzzles)



Implications for Proofs of Quantumness (PoQ)

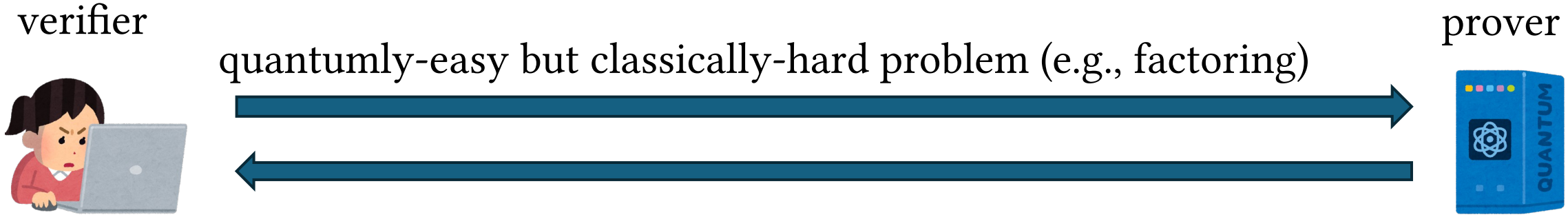
PoQM generalizes proofs of quantumness (PoQ).

$\Longrightarrow$ Studying PoQM gives us some implications to PoQ.

PoQM captures genuinely-interactive PoQ

Two types of PoQ

trivial PoQ :

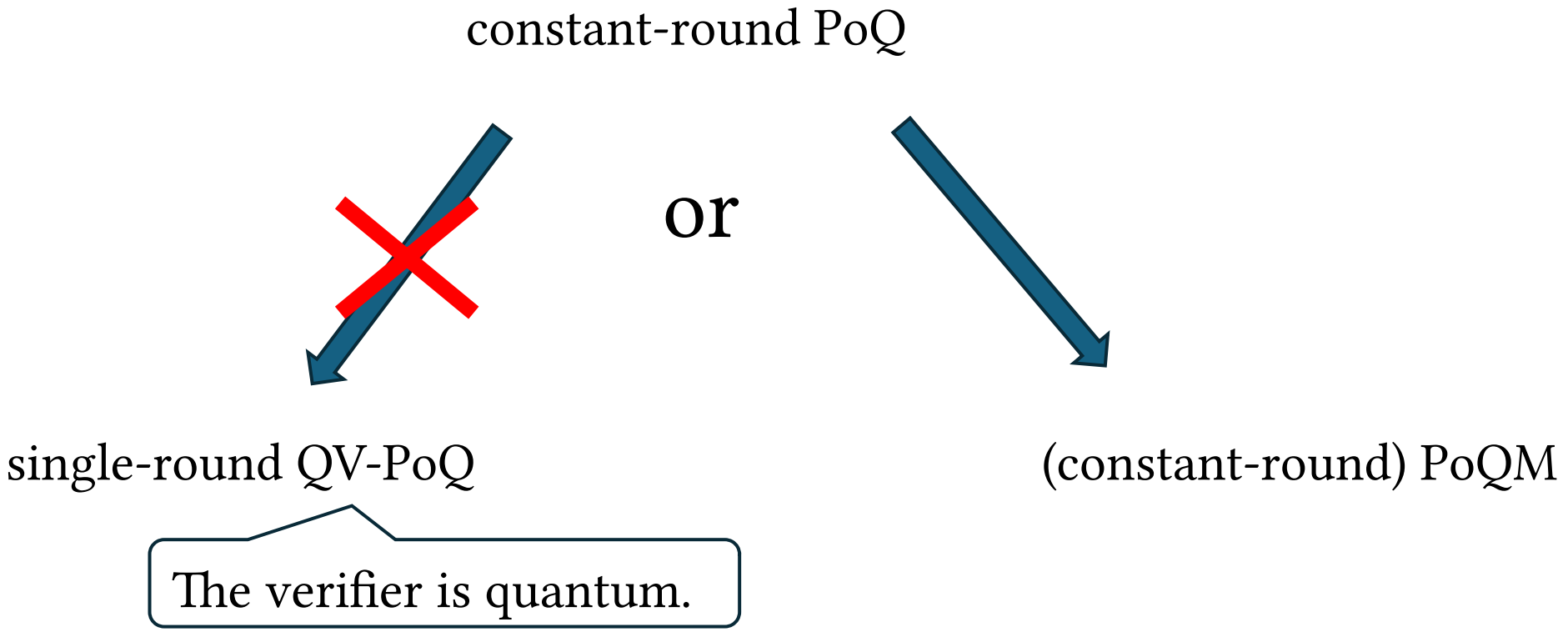


Some PoQ protocols, such as that use the rewinding [Brakerski et al'21], are "genuinely-interactive".

The definition of PoQ includes both types of PoQ and therefore cannot distinguish between them.

Can we capture only genuinely-interactive PoQ?

PoQM capture genuinely-interactive PoQ



inefficient-verifier PoQ (IV-PoQ) based on a quantumly-secure falsifiable assumption imply inefficient-verifier PoQM (IV-PoQM)

IV-PoQ with black-box reduction

In many constructions:

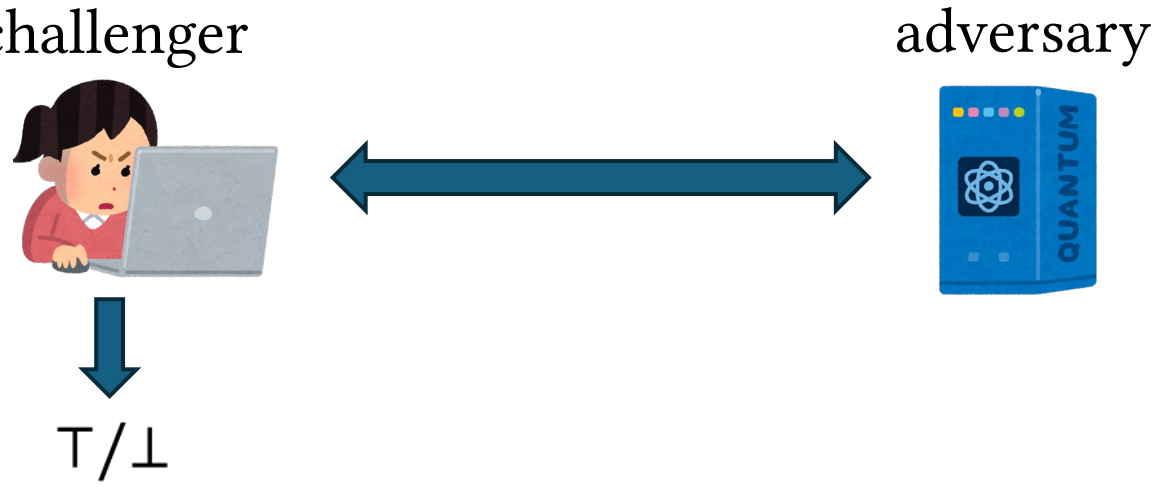
Assumption ⟹ (IV-)PoQ

black-box way

capture a wide class of assumptions : OWFs, PKE, OT, LWE, iO, etc.

The verifier's final computation to make the decision may be inefficient.

Falsifiable assumptions : adversary breaks the assumption if the challenger accepts with high probability.



IV-PoQM from IV-PoQ

constant-round IV-PoQ that are black-box constructed from a quantumly-secure falsifiable assumption



constant-round IV-PoQM

[Morimae-Yamakawa'24]

collision-resistant hash



constant-round IV-PoQ



IV-PoQM

Thank you for listening!